

Kinetics Final Assignment

Haniyeh Zahra Budaqi
haniyehz.budaqi55@sharif.edu

February 4, 2026

Problem One

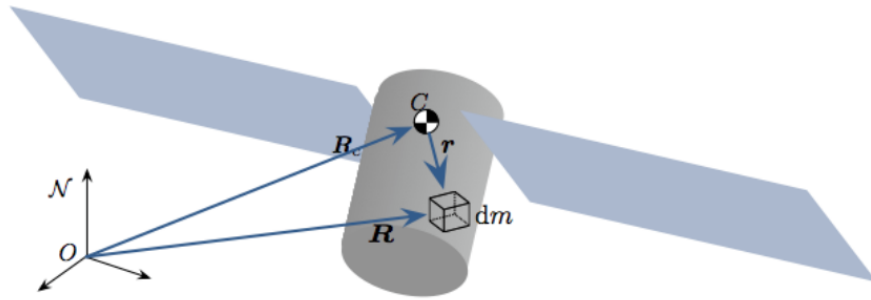


Figure 1: Illustration of Rigid Spacecraft System

Consider the rigid spacecraft system B shown in Fig 1.

- Express the angular momentum vector $\mathbf{H}_c$ about the spacecraft center of mass as an integral over the body B .
- Derive the inertial derivative of the $\mathbf{H}_c$ integral expression. Prove that this is equal to the torque acting about the spacecraft center of mass. Make use of the fact up front that this spacecraft is rigid.
- Starting with the general kinetic energy expression $T = \frac{1}{2} \int_B \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} dm$, derive the kinetic energy expression that expresses it in terms of the energy of the center of mass and the rotational energy about the center of mass. Make use of the fact up front that this spacecraft is rigid. (Hint: recall the vector identity $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$)

Answer

a) From the picture, we know that we can express the angular momentum about the center $\mathbf{H}_C$ as:

$$\mathbf{H}_C = \int_B \mathbf{r} \times \dot{\mathbf{r}} dm$$

$\dot{\mathbf{r}}$ is the inertial derivative of the vector $\mathbf{r}$. Calculating it we have:

$$\begin{aligned} \dot{\mathbf{r}} &= {}^{\mathcal{N}}\left(\frac{d\mathbf{r}}{dt}\right) \\ &= {}^{\mathcal{B}}\left(\frac{d\mathbf{r}}{dt}\right) + \boldsymbol{\omega} \times \mathbf{r} \end{aligned}$$

Since we're assuming the spacecraft is rigid, the position of its elements relative to the center of mass will not change and thus, the body-frame derivative of $\mathbf{r}$ is zero:

$${}^{\mathcal{B}}\left(\frac{d\mathbf{r}}{dt}\right) = 0$$

And $\dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r}$. So expressing $\mathbf{H}_C$ as an integral over the body would be:

$$\mathbf{H}_C = \int_B \mathbf{r} \times \dot{\mathbf{r}} dm = \int_B \mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) dm$$

b) The inertial derivative of $\mathbf{H}_C$ would be:

$$\begin{aligned} \dot{\mathbf{H}}_C &= {}^{\mathcal{N}}\frac{d\mathbf{H}_C}{dt} \\ &= {}^{\mathcal{N}}\frac{d}{dt}\left(\int_B \mathbf{r} \times \dot{\mathbf{r}} dm\right) \\ &= \int_B \left({}^{\mathcal{N}}\frac{d\mathbf{r}}{dt} \times \dot{\mathbf{r}} + \mathbf{r} \times {}^{\mathcal{N}}\frac{d\dot{\mathbf{r}}}{dt}\right) dm \\ &= \int_B (\dot{\mathbf{r}} \times \dot{\mathbf{r}} + \mathbf{r} \times \ddot{\mathbf{r}}) dm \\ &= \int_B \mathbf{r} \times \ddot{\mathbf{r}} dm \end{aligned}$$

Earlier we had the definition for the net torque around some inertial point P :

$$\mathbf{L}_P = \int_B \boldsymbol{\sigma} \times \ddot{\mathbf{R}} dm$$

in which $\boldsymbol{\sigma} = \mathbf{R} - \mathbf{R}_P$.

Supposing point P as the center of mass ($\mathbf{R}_P = \mathbf{R}_C$), we'll have $\boldsymbol{\sigma} = \mathbf{R} - \mathbf{R}_C = \mathbf{r}$ and the integral will be:

$$\begin{aligned} \mathbf{L}_C &= \int_B \mathbf{r} \times \ddot{\mathbf{R}} dm \\ &= \int_B \mathbf{r} \times (\ddot{\mathbf{r}} + \ddot{\mathbf{R}}_C) dm \end{aligned}$$

Since the center of mass is an inertial point, $\ddot{\mathbf{R}}_C = 0$ and:

$$\mathbf{L}_C = \int_B \mathbf{r} \times \dot{\mathbf{r}} dm$$

Along with the last part, since $\dot{\mathbf{H}}_C = \int_B \mathbf{r} \times \ddot{\mathbf{r}} dm$ and $\mathbf{L}_C = \int_B \mathbf{r} \times \dot{\mathbf{r}} dm$, we can conclude that:

$$\dot{\mathbf{H}}_C = \mathbf{L}_C$$

c) From the expression $T = \frac{1}{2} \int_B \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} dm$, we can substitute $\dot{\mathbf{R}} = \dot{\mathbf{r}} + \dot{\mathbf{R}}_C$:

$$\begin{aligned} T &= \frac{1}{2} \int_B \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} dm \\ &= \frac{1}{2} \int_B (\dot{\mathbf{r}} + \dot{\mathbf{R}}_C) \cdot (\dot{\mathbf{r}} + \dot{\mathbf{R}}_C) dm \\ &= \frac{1}{2} \int_B (\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} + 2\dot{\mathbf{R}}_C \cdot \dot{\mathbf{r}} + \dot{\mathbf{R}}_C \cdot \dot{\mathbf{R}}_C) dm \\ &= \frac{1}{2} \int_B \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} dm + \dot{\mathbf{R}}_C \cdot \int_B \dot{\mathbf{r}} dm + \frac{1}{2} \dot{\mathbf{R}}_C \cdot \dot{\mathbf{R}}_C \int_B dm \end{aligned}$$

We know that $\int_B dm = M$ so the last integral results in $\frac{1}{2} \dot{\mathbf{R}}_C \cdot \dot{\mathbf{R}}_C \int_B dm = \frac{1}{2} M \dot{\mathbf{R}}_C \cdot \dot{\mathbf{R}}_C$. We also know that $\int_B \mathbf{r} dm = 0$ (since the spacecraft is rigid) and therefore $\int_B \dot{\mathbf{r}} dm = 0$ and the second term vanishes.

For the first term, we can substitute $\dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r}$ and have:

$$\frac{1}{2} \int_B \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} dm = \frac{1}{2} \int_B (\boldsymbol{\omega} \times \mathbf{r}) \cdot (\boldsymbol{\omega} \times \mathbf{r}) dm$$

Using the identity given by the problem, we can rewrite the term inside the integral as:

$$(\boldsymbol{\omega} \times \mathbf{r}) \cdot (\boldsymbol{\omega} \times \mathbf{r}) = \boldsymbol{\omega} \cdot (\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}))$$

The integral will then be:

$$\begin{aligned} \frac{1}{2} \int_B (\boldsymbol{\omega} \times \mathbf{r}) \cdot (\boldsymbol{\omega} \times \mathbf{r}) dm &= \frac{1}{2} \int_B \boldsymbol{\omega} \cdot (\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r})) dm \\ &= \frac{1}{2} \boldsymbol{\omega} \cdot \int_B \mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) dm \\ &= \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{H}_C \end{aligned}$$

In the end, we'll have:

$$T = \frac{1}{2} \int_B \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} dm = \frac{1}{2} M \dot{\mathbf{R}}_C \cdot \dot{\mathbf{R}}_C + \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{H}_C$$

in which, the first term on the right is the translational energy of the object (energy of the center of mass) and the second term on the right is the rotational energy about the center of mass:

$$T_{trans} = \frac{1}{2} M \dot{\mathbf{R}}_C \cdot \dot{\mathbf{R}}_C$$

$$T_{rot} = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{H}_C$$